Project1

Tittle: Railway Gauges

The library offers straightforward methods for importing various file types and converting them into a Data Frame structure.

Here we are importing a CSV file into Pandas and Pandas returns a Data Frame object.

Figure 1:Loading CSV file into Pandas

```
import pandas as pd
import matplotlib.pyplot as plt
```

Using Pandas Data Frame object we can see **head()**, **tail()** functions to display top 5, last 5 rows from imported dataset.

	Year	Broad Gauge	Metre Gauge	Narrow Gauge	Total
0	1964-65	3521	2891	464	6876
1	1965-66	3603	2913	470	6986
2	1966-67	3615	2911	481	7007
3	1967-68	3601	2920	506	7027
4	1968-69	3603	2921	508	7032

Figure 2: Displaying top 5 rows

Some of the Pandas functions are:

- **read_csv()**-It helps to Loads CSV file into a Data Frame
- **iloc[]** -It is used for selecting data by integer position
- **idxmax()**-Returns index of maximum value

```
# Find which year had the maximum installations
max_year_data = df.iloc[[df['Total'].idxmax()]]
print(max_year_data)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Asus\OneDrive\Desktop\Analytics\Sample Project1>  c
Microsoft\WindowsApps\python3.13.exe' 'c:\Users\Asus\.vscode\ex
ers\Asus\OneDrive\Desktop\Analytics\Sample Project1\main.py'
       Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
48  2012-13         6577          379           216   7172
PS C:\Users\Asus\OneDrive\Desktop\Analytics\Sample Project1>
```

Figure 3: Displaying details of highest Total Year

I used Pandas tools to manage the data: read_csv() for loading, fillna() for missing data, and to_numeric() to fix data types. I also checked the data structure using columns and dtypes. For the actual analysis, I filtered rows with loc[], aggregated values using sum()

Matplotlib is a popular Python library used for data visualization. It allows you to create charts and graphs to better understand and present data.

It works like NumPy and pandas.

Some of the Matplotlib functions are:

- plt.plot() - is used to create a line graph.
- plt.bar() - creates a bar chart. ● plt.show() - Displays the plot.

DATA VISUALIZATION

Matplotlib converts raw data into clear visuals, making it simpler to spot patterns and compare figures. As a staple in science and AI, it allows users to build everything from line plots to pie charts to communicate complex findings effectively.

This code is used to visualize the relationship between Year and Number of railway tracks using a bar graph with Matplotlib.

```python
# Plot data using bar chart
df = df.drop('Total', axis=1)
ax = df.plot(x="Year", kind="bar")
plt.xticks(rotation=70)
plt.xlabel('Year')
plt.ylabel('Total')
plt.title('Gauges: Number of railway tracks installed per year')
plt.savefig('rail_gauges.png')
plt.show()
```

Figure 4: Bar Graph Code

The resulting graph shows how the number of railway tracks changes over different years, helping to identify trends such as increase or decrease over time.

RESULTS

The project successfully analyzes railway gauge data using pandas and Matplotlib, and concludes that Broad Gauge dominates the modern railway system with overall increasing track growth.

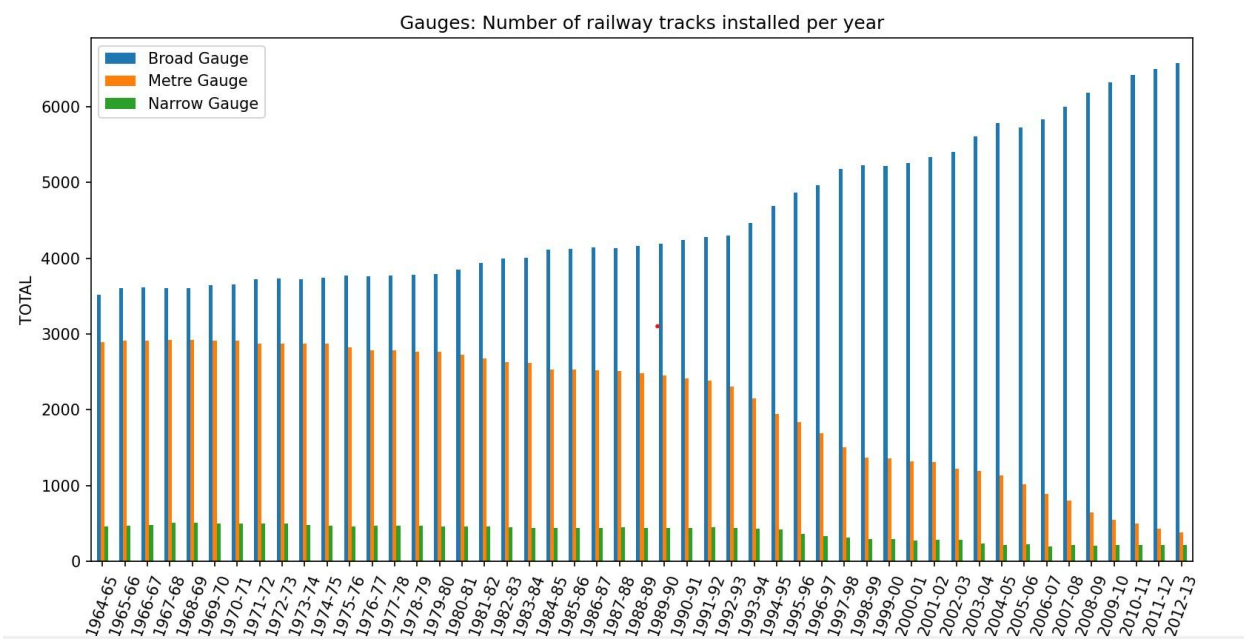

Figure 5: Number of Railway tracks installed per year